

The Reliability Premium: Evidence for Capacity-Constrained Models of Labor Productivity

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Abstract

Most labor productivity measurement rests on a simple substitution: time worked stands in for output produced. This paper argues that optimizing for clock-hours can degrade the very output it is meant to track, and proposes output density — value produced per unit of available human capacity, rather than per clock-hour — as the necessary alternative. It tests this argument across four datasets spanning national economies, individual time-use diaries, and physiological recordings — treating them not as separate studies but as layers of a single constrained system. At the country level, annual work hours and GDP per hour worked move in opposite directions ($r = -0.70$, $N = 43$ countries); countries in the highest-hour quartile produce less than half the hourly output of the lowest-hour quartile. Beneath that macro signal, most of an individual's day is already claimed by structural demands that have almost nothing to do with biology: sleep duration explains less than 2% of how people divide their waking hours between work, commuting, and everything else ($N = 92,949$). The hours that remain are bounded by a recovery buffer — sleep — that depletes at an accelerating rate as work hours rise; the relationship between daily work hours and sleep is convex, not linear, and the tenth hour of work costs more of that buffer than the sixth ($N = 244,026$ American Time Use Survey diaries). At the signal level, heart rate and heart rate variability respond strongly to demand, but the direction of the response depends on whether the demand is psychological or physical, which means capacity is not a fixed quantity that a sensor can read off directly (WESAD and PhysioNet datasets). Read together, these findings describe a system in which clock-hours are not the binding constraint on output — structural and biological capacity are — with consequences for how labor is measured, scheduled, and priced.

KEYWORDS: labor productivity, capacity constraints, time use, heart rate variability, performance measurement, human capital

1. Introduction

The time clock is a nineteenth-century technology, built to solve a specific problem. When work is tied to a machine that runs at a fixed speed for a fixed number of hours, attendance and output are nearly the same thing. Punch in, punch out, and the difference between the two timestamps is a reasonable estimate of what got made.

Very little work still looks like that. But the clock persists — in hourly billing, in attendance-based reviews, in calendar software that treats every open slot as interchangeable, in the headcount math that turns a workforce into a number of "full-time equivalents." The assumption underneath all of it hasn't moved much since 1920: an hour is an hour, and more of them means more output.

Every system has a constraint — the one factor that actually limits what the system can produce, no matter how much effort goes into improving everything else. Spend resources optimizing anything other than that constraint, and total output doesn't move. This paper's argument, stated plainly, is that clock-hours have been treated as the constraint on human productive output — the variable that scheduling, billing, and performance systems are built around — when the evidence below suggests they aren't.

If clock-hours aren't the binding constraint, what is? Underneath the variable everyone measures sits a stack of others that do the actual binding. At the surface, clock-hours and output move in opposite directions: push the lever that billing and scheduling systems are built around, and the system produces less, not more (Section 4.1). That pattern only makes sense once you look beneath it. Most of a person's day is already spoken for before any work happens — claimed by commutes, caregiving, and obligations no scheduling tool sees (Section 4.2). What survives that is bounded by a recovery buffer that drains faster the harder it's pulled; the tenth hour of work costs more than the sixth (Section 4.3). And the body's own moment-to-moment signals, heart rate and its variability, shift with context instead of reporting capacity on a fixed scale (Section 4.4).

One proposition ties these together: that clock-hours are the binding constraint on output. A model built on it expects more hours to mean more output, every time. The data refuses, at every scale tested. What it points to instead is a different denominator — out-

put density, value produced per unit of capacity actually available, rather than per clock hour. No dataset here measures it directly, and that gap is itself part of the finding. What each one supplies is a piece of what such a denominator would have to account for.

2. Data

2.1 Cross-National Panel: OECD, World Bank, Penn World Tables

The country-level analysis draws on a merged panel covering 1991–2023: annual hours worked per worker from the OECD (Total worker-status series, all employed persons), GDP per person employed (PPP-adjusted, 2017 USD) from the World Bank, and GDP per hour worked (PPP-adjusted) from the Penn World Tables. The primary analysis uses the latest common cross-section — 43 countries observed in 2023, one observation per country.

These are national aggregates. A country's average hours figure blends a finance worker in its capital city with a seasonal agricultural laborer in its countryside, and the dataset has no way to separate them. It also has no information on industry mix, labor-market institutions, or how a country's working-time norms got to be what they are. Those gaps matter for how the country-level finding should be read, and they're addressed directly in Section 5.

2.2 American Time Use Survey — Active Workers

The structural-constraint analysis uses a subset of 92,949 ATUS respondents who reported nonzero work hours, and constructs a single derived variable, the **STRUCTURAL WORK LOAD (SWL)**:

$$SWL = (work_hours + commute_hours) / waking_hours$$

where $waking_hours = 24 - sleep_hours$. SWL is the proportion of a day already claimed by structural demands — work, plus the commute it requires — before any biological recovery can occur. Commuting belongs in the numerator for the same reason work does: it is a non-negotiable, structurally imposed draw on the waking day that no amount of ef-

fort or willpower removes. SWL says nothing about how productive that time was, how it felt, or what else the person was capable of; it is strictly a measure of how much of the day was left over once work and commuting were accounted for.

SWL should be read as a measure of how much of the day's recovery time has already been claimed by work, not of cognitive output itself. The numerator that would complete a true capacity measure — output, effort, or cognitive engagement per hour — is not present in this dataset, and this paper does not claim otherwise. What SWL captures is a structural denominator: how much of a day's waking time has already been allocated to work and the commute it requires, and how much smaller a window that leaves for the recovery processes — sleep, but also rest, detachment, and unwinding more broadly (Sonnetag & Fritz, 2007) — that the body needs to return to baseline. A high SWL value describes a *recovery squeeze*: a day in which the non-work window has become compressed, not a measurement of how capable the person currently is. The bridge from recovery squeeze to capacity in the fuller sense runs through a body of occupational health literature connecting time scarcity to physiological wear.

2.3 American Time Use Survey — Full Sample

The biological-constraint analysis uses the American Time Use Survey (ATUS), 2003–2024, with 244,026 person-day observations. ATUS works by asking a representative sample of Americans to reconstruct, hour by hour, what they did the previous day — when they slept, worked, commuted, cared for someone, or did anything else. Every analysis here applies the survey's sampling weights, so the estimates describe the U.S. population, not just the sample.

The key variables are daily sleep hours — the recovery buffer this paper treats as the relevant biological resource — daily work hours, treated as the load placed on that buffer, and daily commute hours, alongside demographic controls: age, sex, employment status, full-time/part-time status, number of children in the household, a weekend indicator, and survey year.

A single diary day cannot show chronic patterns — what someone's sleep looked like over a stressful month, say. It can only show what one day looked like. That limitation is real, and it's carried forward into Section 5.

2.4 HRV Datasets: WESAD and PhysioNet

The fourth analysis draws on two physiological datasets that were never designed to be compared to each other, which is part of the point.

WESAD is a controlled lab dataset: 15 participants, ECG-monitored, moved through three states — resting baseline, induced psychological stress (via a standard social-stress protocol), and a pleasant-video "amusement" condition. From each condition, three features were extracted: RMSSD and SDNN (two common heart rate variability metrics) and mean heart rate, with 15 segments per condition.

PHYSIONET'S WEARABLE DATASET captures the same family of metrics — derived from inter-beat interval data rather than ECG — across three physical workload conditions: aerobic exercise (N = 30), anaerobic exercise (N = 31), and a rest/recovery protocol labeled "STRESS" in the source data (N = 37), which represents a physiologically different state from WESAD's psychological stress condition despite the shared label.

Different equipment, different populations, different definitions of "stress." The two datasets are treated here as independent tests of the same question — are these signals consistent across contexts? — not as a combined sample.

3. Methods

3.1 Descriptive Statistics

The headline sleep and work durations use the survey's sampling weights, so they describe the U.S. working population rather than the raw sample. The SWL ratio and its correlation with sleep, by contrast, are reported unweighted: SWL is a within-day allocation ratio, and the unweighted figures describe the active-worker sample directly. Subgroup breakdowns by employment status, age, sex, full-time/part-time status, and weekday/weekend were used to check that the headline patterns weren't artifacts of a single subgroup.

A few numbers anchor the rest of the paper: the weighted mean sleep duration for active workers is 7.91 hours (SD = 1.82), and mean work duration is 7.60 hours (SD = 3.02). SWL, computed unweighted over the 92,949-respondent active-worker sample, has a mean of 0.464, a median of 0.511, and an interquartile range of 0.274 — meaning the middle half of active workers carry a Structural Work Load somewhere between roughly 34% and 61% of their waking day.

3.2 Correlation Analysis

For the country-level and SWL analyses, both Pearson and Spearman correlations were computed. At the country level, $r(\text{annual hours, GDP/hour}) = -0.697$ (Pearson) and -0.778 (Spearman); on an outlier-trimmed sample, -0.798 and -0.800 . For SWL, $r(\text{sleep hours, SWL}) = -0.131$ (Pearson) and -0.095 (Spearman) — several times weaker than the country-level relationship, which is itself part of the finding in Section 4.2.

3.3 Regression Models

COUNTRY LEVEL. Four OLS specifications were estimated for GDP per hour worked as a function of annual hours: linear ($R^2 = 0.486$), quadratic, log-linear ($R^2 = 0.682$), and log-quadratic. The log-linear specification fits best. The quadratic term is not statistically distinguishable from zero ($p = 0.511$) — the relationship doesn't curve, plateau, or reverse anywhere in the observed range.

STRUCTURAL WORK LOAD (THE STRUCTURAL CONSTRAINT). Two families of models were estimated on the 92,949-person active-worker sample with different dependent variables. Models predicting raw daily work hours from sleep hours reach $R^2 \approx 0.118$. Models predicting SWL — the share of waking time claimed by work and commuting — from sleep hours reach $R^2 \approx 0.017$ – 0.019 . The gap between these two numbers is the central result of Section 4.2.

INDIVIDUAL LEVEL (THE BIOLOGICAL BUFFER). The primary specification regresses sleep hours on work hours plus controls (commute time, age and age², sex, employment status, full/part-time status, number of children, weekend indicator, survey year), $N = 244,026$, $R^2 = 0.149$. The work-hours coefficient is -0.177 ($p < 0.001$). A second specification adds a quadratic work-hours term to test for nonlinearity; marginal effects from this model locate inflection points at approximately 6.86 and 9.80 daily work hours.

3.4 Effect Size Analysis

For the HRV datasets, Cohen's d was computed for all pairwise condition comparisons, alongside Mann-Whitney U tests (given WESAD's small per-condition N) and Student's t -tests where appropriate. PhysioNet heart rate effect sizes were computed from reported group means and standard deviations (aerobic: $M = 106.0$, $SD = 17.0$, $N = 30$; anaerobic: $M = 114.8$, $SD = 18.7$, $N = 31$; rest/recovery "STRESS": $M = 78.8$, $SD = 12.8$, $N = 37$), yielding $d = 1.84$ (aerobic vs. rest) and $d = 2.29$ (anaerobic vs. rest).

4. Findings

4.1 The Symptom: Hours and Output Move in Opposite Directions

Across 43 countries, pushing annual work hours up doesn't raise GDP per hour worked — it erodes it. The relationship is negative ($r = -0.697$), gets stronger rather than weaker once outlier economies are trimmed (-0.798), and holds under rank-based correlation too (Spearman $r = -0.778$).

This isn't a marginal effect. Countries in the lowest quartile for annual hours average around 1,441 hours per year — the equivalent of roughly 28 hours a week, year-round — and produce \$84.33 of output for every hour worked. Countries in the highest quartile average 2,013 hours per year, close to a 39-hour week with little time off, and produce \$34.86 per hour. The high-hour countries put in about 40% more time and get back less than half the output per hour.

The log-linear specification fits best ($R^2 = 0.682$), and the curve never bends back. There's no point in the data where working more starts paying off, and no point where the penalty stops accruing — the quadratic term is statistically indistinguishable from zero. Each additional annual hour costs roughly the same amount of hourly output as the one before it. The line just keeps going down.

This is the false-constraint signature. If clock-hours were the lever that determined output, pulling it harder should produce more, or at worst nothing. Instead it produces less, consistently, across the full range of the data. Something absorbs the cost of those extra

hours, and it isn't showing up in the output column.

4.2 The Structural Constraint: Where the Day Already Goes

Some of it is absorbed before work even begins — by how much of the day is already spoken for.

Among 92,949 working adults, the average Structural Work Load is 46% of waking hours — but that figure ranges from roughly 34% to 61% between the 25th and 75th percentiles. Two people with the same number of waking hours can operate under radically different Structural Work Loads: radically different amounts of their day already claimed before anything else happens.

The obvious candidate for explaining that difference is biological: maybe people who sleep less simply have more hours available, and use them. Sleep duration does predict about 12% of the variation in raw work hours ($R^2 = 0.118$) — and a meaningful share of that is mechanical, since sleeping longer leaves fewer hours in the day during which work could happen at all. But the more direct test is SWL itself — the share of waking time claimed by work and commuting, rather than raw hours — and here sleep's explanatory power collapses to under 2% ($R^2 \approx 0.018$).

That collapse is the structural constraint itself. The variable that sleep trackers, wellness apps, and "recharge" framing are built around — biological recovery state — accounts for almost none of how tightly a person's day is packed. People sleeping less than 7 hours allocate about 3 percentage points more of their waking time to work and commuting (49% versus 46%) than people sleeping 7 or more hours — but they also work fewer total hours, which is consistent with their days being compressed by something other than work itself: commutes, caregiving, second jobs, schedules they don't control.

A recovery squeeze that persists does real damage. The body keeps a running tab. A substantial occupational health literature connects sustained time scarcity to allostatic load — the cumulative physiological wear that builds when stress-response systems fire again and again without enough time to return to baseline (McEwen & Stellar, 1993). None of it resets on its own. The damage surfaces downstream, on a dose-response curve: longer hours, more wear, including incident glucose intolerance (Lee et al., 2022) and elevated allostatic load indices accumulated across a working career (Cuitún Coronado et al.,

2018). The pathway runs straight through the window SWL measures. When work claims more of the waking day, the time the body has to reset shrinks — and the cost compounds.

That window has an empirically measurable floor. When a worker's daily rest period — the off-work time between leaving work and returning to it — drops below roughly 11 to 12 hours, recovery markers deteriorate sharply: workers below that threshold report close to double the rate of psychological distress, along with more non-restorative sleep and more fatigue, than workers with longer windows (Tsuchiya et al., 2017). Short inter-shift rest predicts elevated sleep and health problems across the broader shift-work literature as well (Vedaa et al., 2016), and the European Union writes the same threshold into law as a minimum 11-hour daily rest entitlement. SWL indexes a closely related quantity — the share of waking time work claims — though, as Section 5 notes, it has not itself been benchmarked against these outcomes.

The macro pattern in Section 4.1 stops being a mystery right here. A compressed recovery window feeds allostatic load. Allostatic load degrades the cognitive resources — attention, working memory, executive control — that sustained output depends on (Boksem & Tops, 2008). So a population working longer hours, with proportionally less recovery, should produce less per hour, not more. That is the pattern in Section 4.1, seen from the inside.

The recovery window itself is carved out by structure, not biology. How compressed that window gets is what determines what's left for the one buffer that has to absorb everything else: sleep.

4.3 The Biological Constraint: Sleep as a Buffer Under Load

"A typical day" doesn't exist. Of 244,026 ATUS diary days, 62.8% include zero hours of work. Not low — zero. Most of the days in this dataset aren't workdays at all, which means any model built around a "standard 8-hour day" is describing a minority case.

On the days when people do work, each additional hour draws down the sleep buffer by about 11 minutes on average (coefficient = -0.177, $p < 0.001$), holding commute time, age, sex, employment status, number of children, and weekend status constant. That's a real, statistically solid number — but it's an average, and the average hides the part that mat-

ters. The full model, including every demographic control available, accounts for only 15% of the night-to-night variation in sleep. The other 85% is being driven by something this model can't see.

What the model *can* see is the shape of the draw-down, and the shape isn't flat. Below about 6.9 hours of work in a day, an additional hour barely touches the buffer. Past that point, each additional hour draws down more sleep than the one before it. Past about 9.8 hours, the draw-down accelerates sharply. The tenth hour of work isn't the same unit as the sixth — it's pulled from a buffer that's already most of the way empty, which is what "fragile to disruption" looks like in the data: little capacity left to absorb whatever the next day brings.

One more piece, smaller but telling: weekends restore about 20 minutes of this buffer on average, independent of how much work happened. The calendar itself — not just the workload on it — shapes how much the buffer gets to refill.

Sleep is a buffer this paper can measure directly, start to finish. Not every buffer leaves that kind of paper trail.

4.4 The Signal-Level Reality: What the Body Can and Can't Tell You

Heart rate and heart rate variability are marketed as the signals that catch capacity being drawn down as it happens — in real time, without waiting for someone to reconstruct their day from memory the next morning.

Heart rate turns out to be a strong signal — strong enough that its strength becomes the problem. In WESAD, heart rate rose from 78 bpm at rest to 101 bpm under psychological stress ($d = 1.21$, $p < 0.001$), and the same gap appeared against the "amusement" condition ($d = 1.44$). In PhysioNet, heart rate during physical exertion (106–115 bpm) was far higher than during the dataset's rest/recovery condition (79 bpm; $d = 1.84$ to 2.29). Both datasets show large, unambiguous effects.

But the two effects point in different directions for different reasons. In WESAD, an elevated heart rate signals psychological load. In PhysioNet, an elevated heart rate signals physical exertion. The number is the same kind of number — beats per minute — but what it's reporting on depends entirely on what's happening to the person wearing the sensor.

Heart rate variability, the metric usually marketed as the more sophisticated capacity signal, performed worse on both counts. RMSSD and SDNN showed only modest and often non-significant differences across conditions in either dataset, and neither metric reliably distinguished between types of physical effort — aerobic versus anaerobic exercise produced essentially the same RMSSD signature ($d = 0.07$).

None of this makes these signals useless. It means capacity isn't a fixed number waiting to be read off a sensor — a heart rate reading means something only once you know what's producing it: psychological pressure, physical exertion, or something else entirely. A system that reads "elevated heart rate" as "low capacity," full stop, will be wrong about as often as it's right.

5. Limitations

These are observational patterns, not controlled experiments, and any single one of them admits more than one causal reading. The biological mechanism is the exception: it runs at the level of individual physiology and holds regardless of what national economies are doing. Here is where each result could break.

THE COUNTRY-LEVEL RELATIONSHIP (4.1) is consistent with the long-standing finding in labor economics that hours and hourly output move inversely (Pencavel, 2015; Collewet & Sauermann, 2017), but the direction of causation is genuinely unsettled. High-income countries may work fewer hours *because* they can afford to, not the other way around — wealth could be the cause and shorter hours the effect, which would produce the same correlation with the opposite causal story. The analysis also has no way to separate a country's hours-productivity relationship from its industry mix, its labor-market institutions, or its history; any of these could be doing some or all of the work the hours variable appears to be doing. The dataset's panel structure (1991–2023) was not used here, and a within-country, fixed-effects version of this analysis is the most direct way to narrow this gap — see Section 7.

This reverse-causality concern, however, doesn't fully neutralize the macro finding. Sections 4.2 and 4.3 describe a mechanism — compressed recovery windows accumulating into allostatic load — that operates at the level of individual physiology, independent of a country's wealth or labor-market institutions. If that mechanism holds even partially, it predicts the same direction of correlation through a second pathway: populations whose average workday leaves less room for recovery would be expected to show suppressed hourly output regardless of why their hours are high. What stays unresolved is the mixture — how much of Section 4.1's pattern each pathway accounts for. What doesn't need resolving is whether the biological pathway exists at all. It does; Sections 4.2 and 4.3 trace it. The macro finding does not stand or fall on the reverse-causality story being false.

THE STRUCTURAL-CONSTRAINT FINDING (4.2) compares different people to each other at one point in time, not the same person across different conditions. Some of what looks like a relationship between sleep and time allocation could reflect who tends to face heavier structural demands in the first place, rather than sleep driving those demands. SWL itself, by construction, measures how a day was divided — not how productive any part of it was, how the person felt, or what they were capable of doing differently. A person with a high SWL value might be thriving or might be barely keeping up; the variable can't distinguish between the two, and treating it as a stand-in for wellbeing or output would be a mistake the data don't support.

More fundamentally, SWL is a structural-allocation ratio, not a validated measure of biological or cognitive capacity: it has not been benchmarked against output, cognitive performance, or any physiological marker of fatigue. The allostatic-load literature introduced in Section 4.2 establishes that a mechanism *could* connect recovery-window compression to declining capacity — it does not establish that SWL, in this dataset, tracks that mechanism. Closing that gap — pairing time-allocation data with biomarkers or performance outcomes — is the validation this measure most needs.

THE BIOLOGICAL-CONSTRAINT FINDING (4.3) comes from single-day diaries. A diary can show what one day looked like; it cannot show whether someone has been running a sleep deficit for a month, or whether the recorded day was unusual. Self-reported time use is also subject to rounding and recall error in both directions.

THE SIGNAL-LEVEL FINDINGS (4.4) rest on small samples, especially WESAD's 15 segments per condition — enough to detect the large effects reported here, but not enough to rule out smaller effects that didn't reach significance. Both datasets were collected in controlled settings that may not resemble how these signals behave during a normal work-day, and the two datasets use different measurement technology (ECG vs. inter-beat interval), so their numeric values aren't directly comparable even where the underlying metric is the same. Neither dataset links physiological signals to any actual measure of output or performance — the claim that these signals carry information about capacity rests on their sensitivity to demand, not on a demonstrated connection to what someone produced.

ACROSS ALL FOUR ANALYSES, the datasets cannot be merged into a single model — the case made here is that four independent analyses point the same direction, not that they form one integrated dataset. SWL in particular has not been validated against any external measure of output, and the broader claim — that clock-hours are not the binding constraint on productive capacity — is a description of a pattern, not a finished theory of what should replace it.

6. Implications

6.1 Output Density as the Real Denominator

Hours worked is the denominator in most productivity measurement, from national accounts down to a single team's sprint velocity. Section 4.1 shows that this denominator moves in the opposite direction from the numerator it's meant to normalize — at the country level, more of the denominator is associated with less output per unit of it. A system that tries to increase productivity by increasing hours is, on this evidence, optimizing a variable that isn't the constraint.

The pattern echoes a body of work on presenteeism — the gap between being present and being productive (Hemp, 2004; Aronsson & Gustafsson, 2005). The data here don't measure presenteeism directly, but they describe the shape it would have to take: hours that

don't convert into output at a constant rate, with the rate of conversion getting worse, not better, as hours accumulate.

The alternative is output density: what gets produced per unit of capacity that's *actually available*, rather than per clock hour. Two findings make this framing necessary rather than merely appealing. Most of a day's available capacity is claimed by structure before work even begins, and that structure varies enormously between people for reasons that have nothing to do with effort. What's left is bounded by a buffer that depletes faster the more it's drawn on. A denominator that ignores both is measuring something — just not the thing it claims to measure.

What that looks like in practice depends on the role, but the move is always the same: put available capacity in the denominator, not hours on a clock. Take a software engineer. The hours-based version counts a 40-hour week; the density version counts merged, reviewed work against the focus blocks that were genuinely free of meetings and on-call duty — often a fraction of those 40 hours, and a fraction that varies wildly week to week. A sales development rep: not dials logged, but qualified opportunities created per hour of actual selling time, after the CRM upkeep, territory travel, and pipeline admin that the job structurally imposes are subtracted from the denominator. A medical resident: decisions and procedures handled per hour worked *inside a defensible fatigue envelope*, where the hour at the end of a 28-hour shift is scored as the different unit Section 4.3 shows it to be, not folded into the same hourly average as the first. None of these is the finished metric. Each one just moves the same quantity — capacity actually available — into the denominator, and each is buildable now from data these fields already keep.

The problem runs deeper than measurement, too. If compressed recovery windows accumulate into allostatic load, and allostatic load erodes the cognitive resources that output depends on, then the denominator problem and the macro pattern in Section 4.1 are two views of the same underlying process — one a measurement artifact, the other its physiological cause. This paper can't separate those two readings, but a useful denominator would have to account for both.

6.2 Why Adaptive Scheduling Tools Keep Missing the Real Constraint

Open any scheduling tool — including the current generation of AI assistants that promise to optimize a calendar around when someone has "energy" — and the model underneath is the one this paper has been dismantling: every open slot treated as identical, "free time" defined as whatever's left over after fixed commitments get placed. The "wellness-aware" version of these tools adjusts schedules using sleep or recovery data — but Section 4.2 already showed what that variable can and can't explain. Sleep accounts for under 2% of how a day gets divided. The other 98% is fixed meetings, caregiving windows, second jobs, schedules people don't control: constraints the tool usually can't see and isn't even asking about. A system that reads a sleep tracker and calls itself adaptive is optimizing the 2% and ignoring the 98%.

Even the slack that's visible isn't uniform, which is the error the sleep-buffer finding makes hard to ignore. A free hour at 9 a.m. and a free hour at 9 p.m., after nine hours of work, are not the same unit of capacity — the second is drawn from a buffer that's nearly empty. A tool using a flat cost function will schedule a demanding task into the most depleted hour of someone's day and log it as a successfully filled slot.

There's a final complication: a wearable signal read without context — the same elevated heart rate meaning focus in one moment and depletion in the next — gives an algorithm exactly enough information to be confidently wrong. None of this is an argument for more sensors. It's an argument for retiring the assumption underneath all of them: that the calendar is the thing to optimize, when the calendar was never the constraint.

6.3 Performance Management Built on the Wrong Variable

Sections 4.2 and 4.3 together describe two people with identical clock-hours who are not operating under remotely similar conditions. One person's day has structural slack in it (4.2); the other's doesn't, for reasons — commute, caregiving, a schedule they didn't design — that have nothing to do with effort or ability. And even between two people whose days look similarly packed, one may be drawing from a sleep buffer that's nearly full while the other's is nearly empty (4.3). A performance system that measures both pairs against the same hourly benchmark is comparing different things and calling them the same thing.

The *reliability premium* emerges precisely in this gap: when two people produce the same output, but one does it from a position of structural slack and a well-stocked buffer, and the other does it while absorbing constraints and running down a buffer the system doesn't see, the second person's output reflects something the first person's doesn't. Current performance metrics don't distinguish between these two cases — they can't, because they're built on a variable, clock hours, that doesn't carry this information. A constraint-adjusted performance measure — output relative to *structurally and biologically available* capacity, not output relative to hours on a calendar — would.

6.4 The Commercial Case for Measuring Reliability

If clock-hours are a noisy proxy for capacity, then any system priced on hours — billing, compensation, workforce planning, hiring — is carrying that noise as a cost, whether or not it's visible on a balance sheet.

A worker who delivers consistent output while absorbing real variability — in schedule, in physiological state, in competing demands — is a different asset than a worker whose output depends on everything staying stable. The first worker is more valuable in a world that doesn't stay stable, which is the world most organizations actually operate in. Current measurement systems don't distinguish between these two workers; if anything, the worker whose conditions are visibly less stable often looks *worse* on hours-based metrics, not better.

The practical openings are in hiring (identifying people who maintain output under real-world variability, not just in ideal conditions), compensation (rewarding demonstrated reliability rather than hours logged), and workforce planning (building plans around what people can sustainably produce under realistic constraints, rather than around an idealized 40-hour week that Section 4.3 suggests doesn't describe most days anyway).

For anyone evaluating a workforce from the outside — an investor, an analyst, an acquirer — the implication is that hours-based productivity metrics may carry systematically different noise across organizations, depending on how much structural variability their workers absorb. A firm with high output density and high structural constraint might look mediocre on a metric like revenue-per-FTE while actually outperforming a firm with lower constraint and the same revenue. This paper doesn't size that gap. It argues the gap is worth measuring.

None of this is an argument against hours as a unit — they're administratively convenient and not going anywhere. It's an argument that hours alone are an incomplete denominator, and that the size of the incompleteness is itself a number worth knowing.

7. Next Steps

PANEL ANALYSIS OF THE COUNTRY-LEVEL DATA. The merged dataset already spans 1991–2023; this analysis used only the latest cross-section. A fixed-effects panel model — tracking each country against itself over time, rather than against other countries — would directly address the reverse-causality concern in Section 5: if a country's productivity rises as its hours fall over time, that's a different and stronger claim than a cross-sectional correlation can support.

FOLLOWING INDIVIDUALS, NOT JUST DAYS. ATUS doesn't track the same person over time, but synthetic cohort methods — following demographic groups across survey years — could show how SWL shifts with policy changes (paid leave expansions, right-to-disconnect laws) or economic conditions, which a single-day snapshot can't.

VALIDATING SWL AGAINST SOMETHING REAL. The work-hours/waking-hours ratio in Section 4.2 is a behavioral measure, not an outcome measure. The allostatic-load literature cited in Section 4.2 suggests the most direct validation targets: pairing time-allocation data with biomarkers of cumulative physiological wear (e.g., cortisol patterns, cardiovascular and metabolic markers comprising an allostatic load index) or with direct measures of cognitive performance would test whether a compressed recovery window actually predicts the biological and cognitive costs SWL is meant to stand in for — rather than only how a day was divided. Earnings, retention, and wellbeing outcomes would be a useful complementary check.

TAKING THE PHYSIOLOGICAL WORK OUT OF THE LAB. WESAD and PhysioNet establish that signals respond to demand inconsistently across contexts; they can't show what that looks like during an actual workday. A study collecting continuous physiological data from people doing real work, paired with in-the-moment reports of what kind of demand

they're under, would test whether Section 4.4's findings hold outside a controlled setting — and would be the first step toward the contextual model that 4.4 says these signals need.

BUILDING THE STRUCTURAL MODEL THAT 4.2 IS MISSING. The under-2% figure in Section 4.2 is most useful as a gap to fill. ATUS already contains many of the relevant variables — household composition, full-/part-time status, day-of-week patterns, multiple-jobholding — that plausibly account for the other 98%. A model that includes them alongside sleep would show how much of daily SWL is actually explainable, and by what.

8. Conclusion

The time clock measured something real once. When a worker and a machine ran at the same fixed speed, the hour was a fair stand-in for the output, and the systems built on that equivalence worked because the equivalence held. It stopped holding. Not because anyone disproved it, but because the work changed underneath it and the unit never moved.

What the four analyses share is a direction, not a dataset. A country adds hours and loses output per hour. A day fills with obligation before work touches it. A buffer drains faster the harder it's pulled. A sensor reports load without reporting what kind. Pull back far enough and these stop reading as separate results. They are one system, measured at four depths, saying the same thing about a unit that no longer tracks what it was built to track.

Output density is the unit that would. It isn't worked out here, and pretending otherwise would be its own version of the clock-hours error — mistaking a clean number for a settled question. But the move it asks for is concrete enough to make now. Stop counting the hours a person was present. Start asking what capacity was actually available to them when they produced what they produced. Every organization that prices, schedules, or evaluates labor already runs that calculation a thousand times a day, with hours sitting in the denominator by default. The claim of this paper is that the default is wrong — and

that the size of the error is not a rounding detail but a quantity worth naming, measuring, and pricing. The first system that learns to see it will be looking at something its competitors still can't.

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All analyses were conducted in Python. Because this paper synthesizes data across four distinct methodologies and scales, the replication code and data access instructions are maintained in separate, specialized environments. A centralized directory linking to all four codebases, along with the final figures and tables, is available at: <https://github.com/ReynaGowie/reliability-premium-synthesis>